Centre Number				Examination Number											

MINISTRY OF EDUCATION

CHINGOLA DISTRICT

NABOONGA SECONDARY SCHOOL

Design and technology Grade 12 End of term 1 test - 2024

Paper 1 (Theory & Design)

6045/1

Tuesday **22nd APRIL 2024**

Additional Materials
A3 Drawing paper
Drawing instruments

Time: 2 hours 30 minutes **Marks: 110**

Instructions to Candidates

1. Write the **centre number and your examination number** on every page of this question paper and on the separate **Answer Booklet** provided.
2. If working is needed for any question, it must be shown in the spaces provided in the question paper and Answer Booklet.
3. There are **two** sections in this paper, **A** and **B**
   - **(i)** Answer all questions in section A in the spaces provided in this question paper
   - **(ii)** Section B has three parts **I, II** and **III**; Part I is compulsory.
   - **(iii)** Choose one question from part **II** and write your answers in the separate Answer Booklets provided.
   - **(iv)** Choose one question from Part III and use the A3 drawing paper(insert) provided.
   - **(v)** Enter the numbers of questions you have answered from Section B Parts II and III in the grid provided.
4. W here a freehand sketch is required, it must be drawn to half full size with good line work.
5. At the end of the examination, fasten the Answer Booklet and drawing paper used securely to the question paper.
6. Use your own discretion where measurements are not provided.

Information for candidates

1. The number of marks is given in brackets [ ] at the end of each question or part question
2. Use of color is allowed
3. All dimensions are in millimetres
4. Cell phones are **not allowed** in this examination room

For Examiner's Use Only			
	Question	Mark Obtained	Examiner's initials
Section A			
Section B			
Part I			
Part II			
Part III			
Total			

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SECTION A. [30 MARKS]
Answer all the questions from this section in the spaces provided in this question paper.
If working is needed for any question, it must be shown in the space below that question.

1. (a) (i) State **two** risks to personal safety when working on a table circular saw.

……
…… [2]

(ii) Mention any precaution that should be taken into for a risk stated in **(a) (i)**

……
…… [1]

(b) **Figure 1.** below shows a plastic tray



Figure 1

(i) Identify the method of plastic moulding used.

…… [1]

(ii) Describe briefly the difference between thermoforming and thermosetting plastic.

……
…… [2]

[Total: 6 marks]

2. (a) **Figure 2** shows the drawing of a pre-finishing tool.

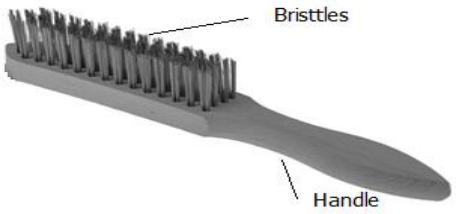


Figure 2

(i) Identify the tool.

…… [1]

(ii) Mention the function of the tool.

…… [1]

(iii) State **one** property which allow bristles to be made from such a material used.

…… [1]

(b) Explain the importance of each of the following in a workroom.

(i) Ventilation: …… [1]

(ii) Lighting: …… [1]

(c) Mention one recommended method of disposing waste generated in the workroom.

…… [1]

[Total: 6 marks]

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3. (a) **The figure** below shows the types of projections used in graphic communication

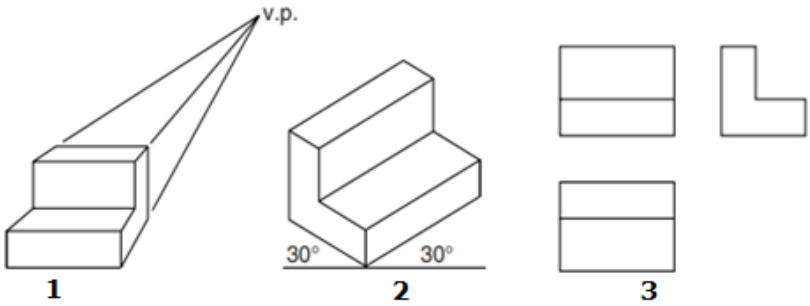


Figure 3

(i) Name the types of projections numbered **1** and **2** shown in **figure 3.**

1. .....
2. ..... [2]

(iii) Explain why projection **type 3** is seen to be the most suitable type of projection for producing working drawings.

.....
..... [1]

b) Chisels are used mostly for making mortises and tenons

(i) Name **two** types of chisels

..... [1]
..... [1]

(ii) State **one** safety rule which should be followed when using chisels.

.....
..... [1]

[Total: 6 marks]

4. (a) **Figure 4** shows the drawing of an electronic circuit diagram

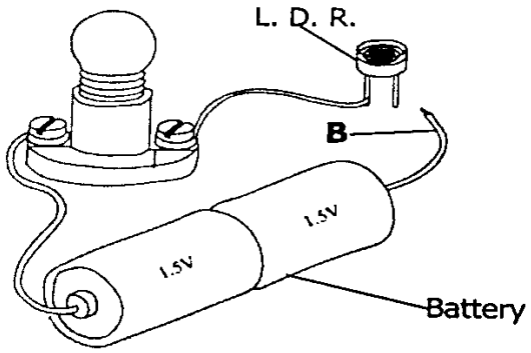


Figure 4

(i) What do the letters **L. D. R.** in the circuit diagram stand for?

..... [1]

(ii) Suggest one appropriate method of joining the wire **B** to the **L. D. R**.

..... [1]

(iii) State the function of a battery in the circuit.

..... [1]

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(b) The A designer wishes to produce a mechanical tin opener using an electric motor and battery. A schematic **diagram 5** is shown below

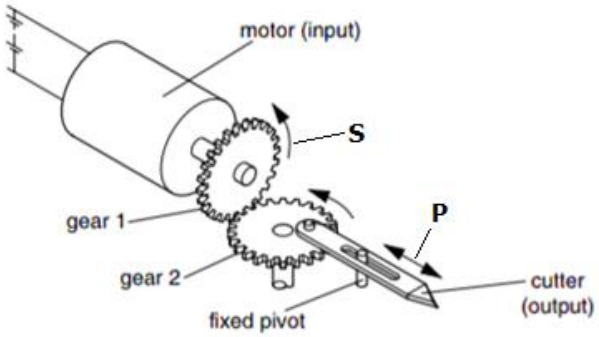


Figure 5.

(i) Identify the type of motion conversion at point **S** and **P.**

P ..... [1]

S ..... [1]

(ii) Mention **one** factor that the designer would have considered when selecting materials for making the above artifact.

..... [1]

[Total: 6 marks]

5. (a) **Figure 6** shows the drawing of a bench tool.

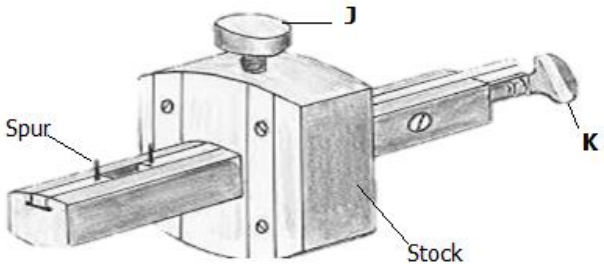


Figure 6.

(i) Identify the tool.

..... [1]

(ii) Name the parts labeled **J** and **K**.

.....

..... [2]

(iii) State the class of tools it belongs to.

..... [1]

(b) What do you understand the following business terms

a) Liquidation: .....

..... [1]

b) Liquidator: .....

..... [1]

[Total: 6 marks]

SECTION B: [80 MARKS]

There are three parts in this section; Part I, II and III. Part I is compulsory. Answer one question from part II and one question from part III. Write your answers in the separate Answer Booklet provided.

Part I: Compulsory

6. Study **figure 7** below obtained in a garage where it is very difficult repairing underneath the car with awkward access and the tools required are not easily at hand.



Figure 7

a) **(i)** Write the situation obtaing in **figure 7.** **[**2]

(ii) Identify the problem from situation in **(a) (i).** [2]

(iii) Derive a design brief from the problem in **(a) ii.** [2]

b) **(i)** Using annotations sketch a pictorial view of **one** possible solution to the situation. [10]

(ii) Mention **two** method of construction you would use in making your solution. [2]

(iii) Apart from mode of operation, suggest four specifications you would consider when designing your artefact. [4]

(c) **(i)** Give **one** reason why exploded views are important in the design process. [1]

(ii) State **two** areas that can be considered when evaluating the artifact. [2]

[Total: 25 marks]

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Part II: Answer one question.

7. (a) Figure. 8 shows an electronic design for a kitchen timer

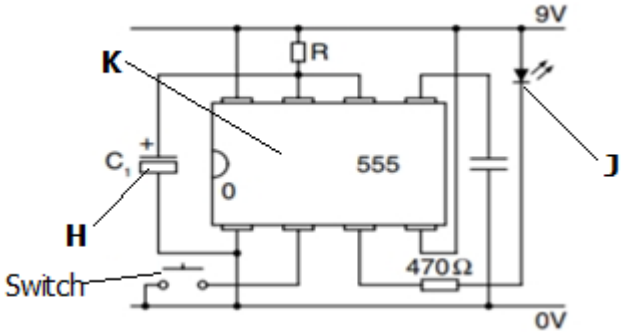


Figure 8

(i) Identify the electronic component labeled **J** and **K** [2]

(ii) State the function of component **H**. [2]

(iii) State the type of switch shown in the drawing. [2]

(iv) A 9V battery powers the circuit. If the protecting resistor for the LED is 470ohms, calculate the current flowing through the LED. [4]

(b) Figure 9 shows a saw used to cut small accurate work pieces.

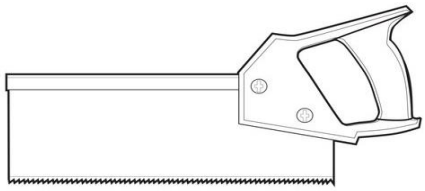


Figure 9

(i) Name the type of saw shown in **figure 19** above. [2]

(ii) State the reason why the teeth of a saw above are 'set'. [2]

(iii) Describe how the saw is designed to cut accurate work. [3]

(c) The figure below shows one of the types of bonds in bricklaying.

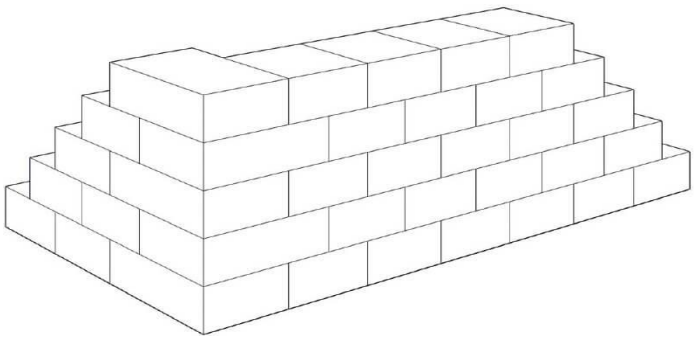


Figure 10

(i) Name the above bond.

(ii) Explain the following building terms

a) Aggregates. [2]

b) Mortar. [2]

(iii) Sketch the following building tools

a) Builders trowel. [2]

b) Builders metal float. [2]

[Total: 25 marks]

8. The framework for a child chair is shown below.

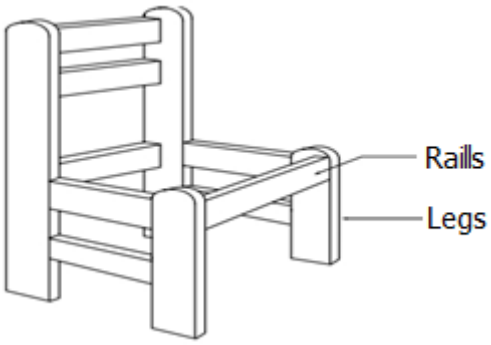


Fig 11.

a) **(i)** Choose a suitable wood material for the chair. [2]
(ii) Give **two** reasons for each, why the materials in (a)(i) are suitable. [2]
(iii) Mention any two tools you would use for fixing the seat to the frame. [2]
b) **(i)** Sketch two different;
1. joints for joining rails and the legs [4]
2. tools to use when constructing the chair. [4]

c) The figure below shows a castor for a shopping trolley. The wheel for the castor is made of a suitable plastic.

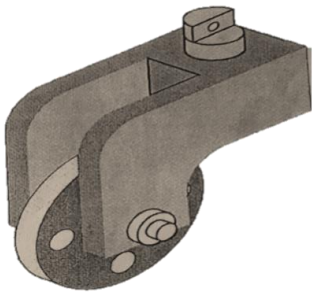


Figure 12.

(i) Give two specific properties of plastic a designer should consider before selecting the material. [2]
(ii) List any **two** ways in which plastics behave when subjected to heat. [2]

d) **(i)** State the principle on conservation of energy. [2]
(ii) List **three** ways in which energy can be transferred. [3]
(iii) State **two** benefits of using solar panels to produce electricity. [2]

[Total: 25 marks]

9. **Figure 13** shows a drawing of a kitchen utensil

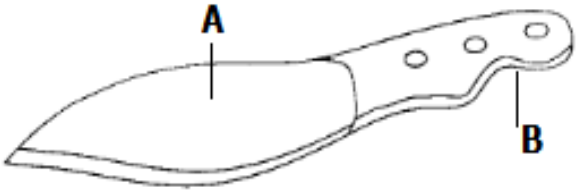


Figure 13

(a) **(i)** Give two properties of material **A** and **B** that makes it suitable for its function. [2]

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(ii) What do you understand the terms;

1. Malleability. [1]
2. Ductility. [1]

(iii) What is meant by the term an **alloy**? [1]

(b) **Figure 14** shows one of equipment used as a method of generating power.



Figure 14

(i) Why should the base frame of equipment in the figure above be made wider. [2]

(ii) State the method used to strengthen the above structure. [1]

(iii) Give one advantage and one disadvantage of generating power using the method in **Figure 14**. [2]

(b) **Figure 15** shows marking out tools.

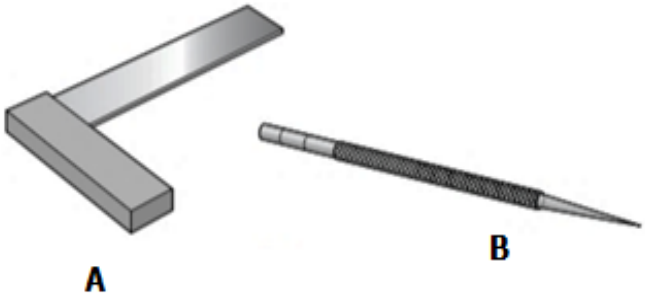


Figure 15

(i) Name the tools **A** and **B** above. [2]

(ii) Explain the function of each tool shown in **figure 15**. [2]

(iii) Mention two personal protective equipment (safety attire) to be worn when welding. [2]

(c) Most businesses start well and fail after some time.

(i) State two reasons why businesses fail. [2]

(ii) Why is government encouraging people to venture into entrepreneurship? [1]

(d) Sketch the following tools and label the parts.

(i) Bevel edged Chisel. [3]

(ii) Hack saw. [3]

[Total: 25 marks]

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Part II: Answer one question on A3 paper provided.

10. Figure 16, represents an outline of a Fish pond at Naboonga secondary school.

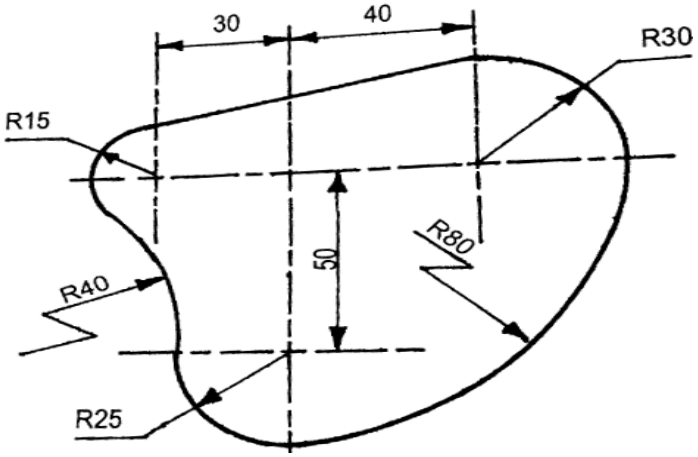


Fig. 16

a) Do not copy, but draw full size the outline of the fish pond. Show clearly all geometrical constructions and points of tangency.

b) Naboonga Boarding school has a canteen that serves meals to borders during lunch and super time. You have been asked to design models that would contain a salt seller and a bottle of spices to be placed on each table in the canteen. Containers could be of any shape as long as they are not cylinders.

(i) Sketch a model for the proposed design in pictorial view.
(ii) Render the model
(iii) Differentiate using sketches by indicating the container for spice and salt.

[Total: 30 marks]

11. A design of a large lamp shaped is to be fixed outside the school dining room.
The shade has the shape of a truncated cylinder of diameter 60. It is made of thin material. A front Elevation of the lamp shade is given in ***figure 17***.

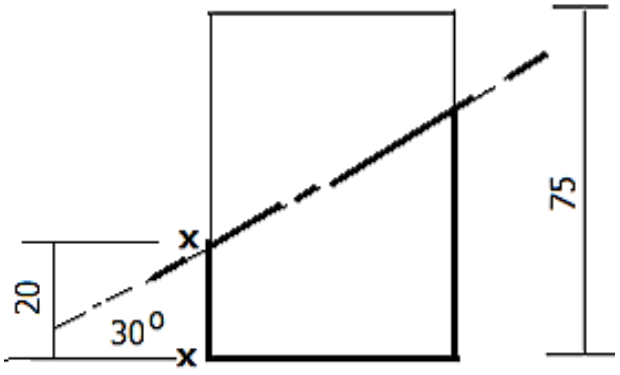


Figure. 17

a) **(i)** Copy the given view and produce the plan. **[6]**
(ii) Determine the pattern of the lamp shade (surface development) before folding it along **X – X.** **[14]**

b) In order to prevent the transmission cholera, the school wants to ensure that learners abide by hygiene regulations.
Design a well rendered sign to illustrate graphically the message **'WASH YOUR 'HANDS"** in concentric circles of diameters 80 and 75 respectively**.** **[10]**

[Total: 30 marks]